A 3nm 125 TOPS/W-29 TFLOPS/W, 90 TOPS/mm^2-17 TFLOPS/mm^2 SRAM-Based INT8 and FP16 Digital-CIM Compiler with Multi-Weight Update/Cycle

Haruki Mori[1], J-M. Hung[1], W-C. Zhao[1], K. Khare[1], B. Crafton[2], H. Ishii[3], C-E. Lee[1], X. Peng[2], X. Sun[2], Y. Fujino[4], C-J. Tsen[1], V. Joshi[5], C-K. Chuang[1], T. Hashizume[4], C-F. Lee[1], T-L. Chou[1], K. Akarvardar[2], S. Adham[5], Y. Wang[1], H. Fujiwara[1], Y-D. Chih[1], Y-H. Chen[1], H-J. Liao[1], T-Y. Jonathan Chang[1]

[1]TSMC, Hsinchu, Taiwan, [2]TSMC, San Jose, CA, [3]TSMC, Osaka, Japan, [4]TSMC, Yokohama, Japan, [5]TSMC, Ottawa, Canada

Abstract

This paper presents a DCIM compiler supporting INT8 and FP16 formats, offering configuration flexibility, high accuracy, and high area/power efficiency. Our 3nm test chip is fully validated and exhibits 124.6 TOPS/W at 0.5V and 90.2 TOPS/mm^2 at 1.1V for INT8 DCIM, and 28.6 TFLOPS/W at 0.5V and 17.4 TFLOPS/mm^2 at 1.1V for FP16 DCIM, respectively.

Introduction

As DNNs support increasingly complex image and language applications including generative AI; the accuracy, flexibility, and power efficiency requirements become more stringent. Although prior DCIM designs demonstrated excellent power efficiency, their flexibility faced challenges [1-2]. This work introduces the first DCIM compiler design, offering flexible configuration, high compute precision, and improved PPA.

Fig. 1 compares two common flows for FP computation. The Alignment First (AF) approach aligns the mantissa bits according to max exponent (EMAX) values before multiplying the weights and activations (XIN). Because the mantissa alignment/truncation are performed upfront, errors from both weight and XIN individually contribute to mantissa product. Accordingly, we observe a visible arithmetic accuracy loss with AF, which had been adopted by prior designs [3-5] using BF format (featuring similar trends as FP format with respect to MF vs. AF). The Multiply First (MF) approach, in contrast, performs mantissa multiplication before the bit alignment. Therefore, it exhibits no accuracy loss for a 22b product (considering 11b mantissa for FP16, including the hidden bit). The plot in Fig. 1 (right) shows the SQNR-based arithmetic accuracy [6] as a function of post-aligned mantissa bitwidth. Considering the MF's SQNR is as high as the FP16 software reference for ≥ 16b (and visibly higher than the AF's SQNR with 22b), in this work, we proceeded with MF flow featuring a mantissa bitwidth of 16b before the accumulation.

INT8 and FP16 DCIM Compilers

Fig. 2 and Fig. 3 provide an overview of INT8 and FP16 DCIM compilers in 3-nm. The INT8 DCIM supports signed/unsigned INT8 for weight and XIN. Each local-MAC (LMAC) includes 16cols x 2 to 18rows SRAM, 8b x 8b multiplier and adder tree. The MAC unit processes INT8 x INT8 parallel MAC with 4,096 bits weight (8b x 64 input x 8 output channels) and 512 bits XIN (8b x 64 input channels) in three-stage MAC pipelining [1]. As illustrated in Fig. 3, FP16 DCIM supports FP16 and signed/unsigned INT8 for weight and XIN. Exponent handling with EXP addition/subtraction, EMAX, mantissa shifter, and format converters are newly adopted in the LMAC and Global-IO (GIO). The FP16 converter handles hidden bit at the interface. The FP16 x FP16 or INT8 x INT8 parallel MAC operations are handled by MAC unit with 8,704 bits weight (17b x 64 input x 8 output channels) and 1,088 bits XIN (17b x 64 input channels). The MAC is implemented with a Booth multiplier to support mantissa alignment. EMAX search receives 384 bits (6b ESUM x 64 input channels) and sends 6b EMAX to LMAC. An EXP subtraction block computes the EXP difference, while a shifter block aligns/truncates mantissa to 16b based on this difference. After conversion to 2's compliment format for accumulation, LMAC outputs 17b mantissa product (1b-sign + 16b-data). GIO includes 18b to 23b adders and FP32 converter, which combines 23b PSUM and 6b EMAX to form a FP32 NOUT. FP16/INT8 modes switch with the FLOATE signal.

Fig. 4 shows the layout floorplan for LMAC and bitcell array. We placed two weight data in vertical direction to mitigate routing congestion due to the adder tree and EMAX search. In order to support compiler function, the number of extra SRAM storage rows is made tunable from 2 to 18. In the largest configuration (64 inputs x 8 outputs channels, 18 SRAM storage rows), SRAM array is organized in 32 banks, and each bank consists of 18 rows and 272 columns. The 272 columns include 17b for FP16, 2 input, and 8 output channels. A 12T, 0.055 μm² bit cell [7] is adopted to enable V_{MIN} < 0.5V.

Fig. 5 shows EMAX search and f_{MAX} comparison between the 1st and the 2nd MAC stages. In the early stages of EXP search, a 4-way comparator is used to minimize the number of gate delay stages. On the other hand, bitwise comparison with a 2-way comparator is adopted for the later stages of EXP search to minimize global wire tracks. The number of gate delay stages between MAC stage1 (4-layer EMAX search + mantissa shifter) and the MAC stage2 (adder tree + FP32 converter) are accordingly balanced. STA results confirmed the difference in total number of gate delay stages is < 5% and f_{MAX} of MAC stage2 is within 7% of MAC stage1 at 0.975V.

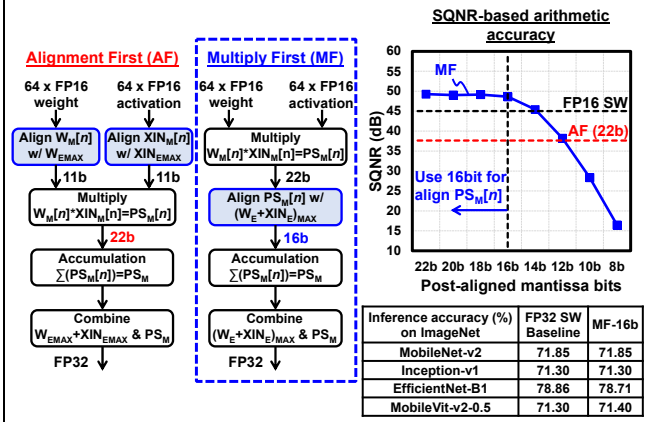
Fig. 6 shows optimization of memory array and adder tree. Prior designs [1-2] updated the weight data for one input channel per cycle (17b across 8 output channels). This limited bandwidth leads to computation stalls, which can be particularly problematic for vector-matrix multiplications (with no weight reuse). To address this issue, we adopt multi-weight update per cycle (e.g., 17b x 2 input channels across 8 output channels), doubling the write bandwidth and halving cycles, enhancing utilization and throughput with negligible area overhead. In addition, a SUM-inversed FA is introduced for 8% better power efficiency compared to polarity-inverted FA, which slows both carry and sum propagations due to the large fan-out, causing MSB side timing gap and glitches.

Fig. 7 illustrates how the tiling is supported in DCIM compiler. Flexible CIM macros are automatically generated by the compiler engine based on the block level leaf cells. We defined three configuration parameters. The number of rows and number of output channels correspond to the word depth and width in the SRAM compilers, respectively. The number of input channels can be handled as in single-port multi-bank (SPMB) design. In the SPMB, reuse of leaf cells is convenient because global routing can be shared. In the DCIM, global adders need extra tie-low leaf cells to maintain global routing since the number of input channels is reduced. We set the range for the extra SRAM storage rows from 2 to 18 with a step size of two rows, the number of output channels from 2 to 8 with a step size of one channel, and the number of input channels from 16 to 64 with a step size of 16 channels.

Chip Implementation and Measurement Results

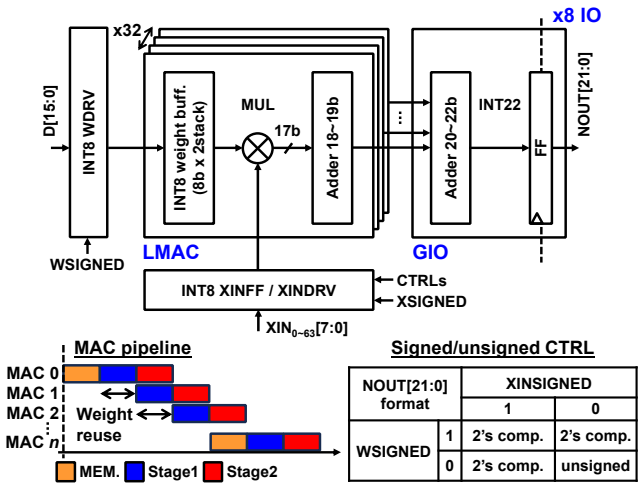
3nm test chip micrograph is shown in Fig.11. In our test chip, CIM BIST and ATPG for DFT are implemented. Fig. 8 summarizes V_{MIN} and f_{MAX}. V_{MIN} is 0.37V and 0.41V for INT8 and FP16 DCIM, respectively, at 160MHz and −40 °C. FP16 V_{MIN} is limited by CIM f_{MAX} during the measurement. We confirmed f_{MAX} with 1.54GHz and 0.72GHz for INT8 and FP16 DCIM, respectively, at 1.1 V. Fig. 9 summarizes the power efficiency: 124.6 TOPS/W (INT8) and 28.6 TFLOPS/W (FP16) at 0.5V, 10% XIN toggle, and no memory access condition. Fig. 10 compares this work with prior art. Our DCIM compiler offers high arithmetic precision and configuration flexibility while maintaining comparable PPA to rigid INT8 DCIM design in [1].

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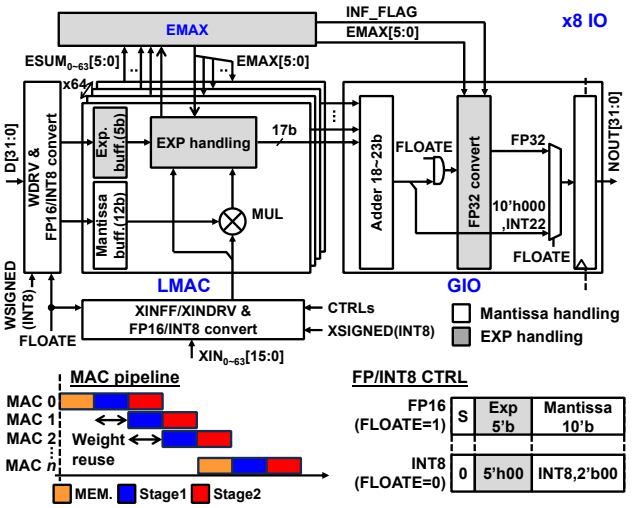
Inference accuracy (%) on ImageNet	FP32 SW Baseline	MF-16b
MobileNet-v2	71.85	71.85
Inception-v1	71.30	71.30
EfficientNet-B1	78.86	78.71
MobileVit-v2-0.5	71.30	71.40

Fig. 1. Left: Alignment First (AF) and Multiply First (MF) flows for DCIM FP16 computation. Right: SQNR-based arithmetic accuracy and inference accuracy comparisons.



NOUT[21:0] format		XINSIGNED	
		1	0
WSIGNED	1	2's comp.	2's comp.
	0	2's comp.	unsigned

Fig. 2. Overview of INT8 DCIM compiler design.



FP16 (FLOATE=1)	S	Exp 5'b	Mantissa 10'b
INT8 (FLOATE=0)	0	5'h00	INT8,2'b00

Fig. 3. Overview of FP16/INT8 DCIM compiler design.

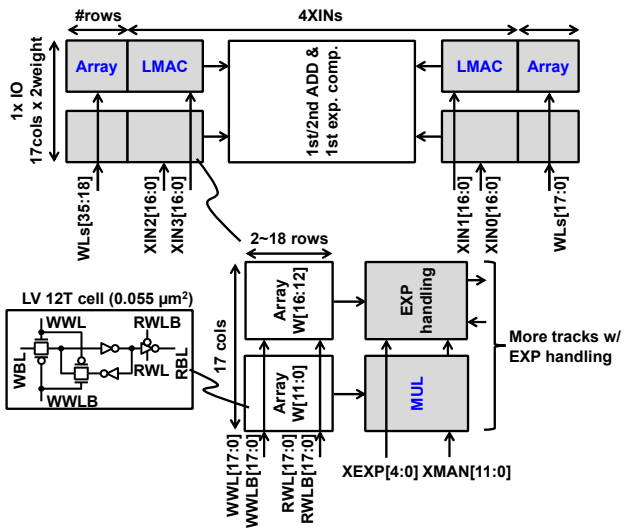


Fig. 4. Layout floorplan for LMAC and bitcell schematic.

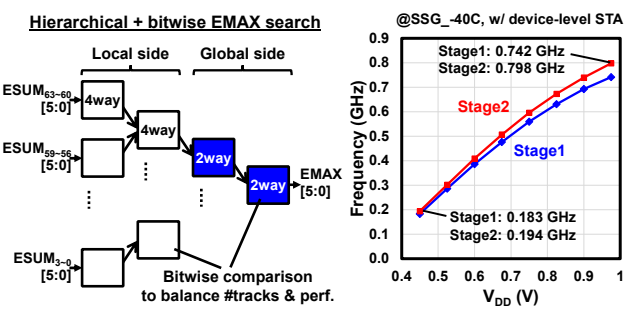
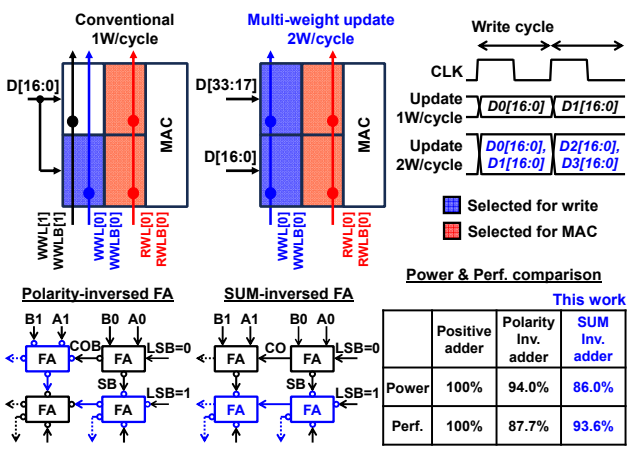


Fig. 5. Maximum exponent search and performance comparison between MAC stage1 and stage2.



	Positive adder	Polarity Inv. adder	SUM Inv. adder
Power	100%	94.0%	86.0%
Perf.	100%	87.7%	93.6%

Fig. 6. Optimization of memory array and adder tree.

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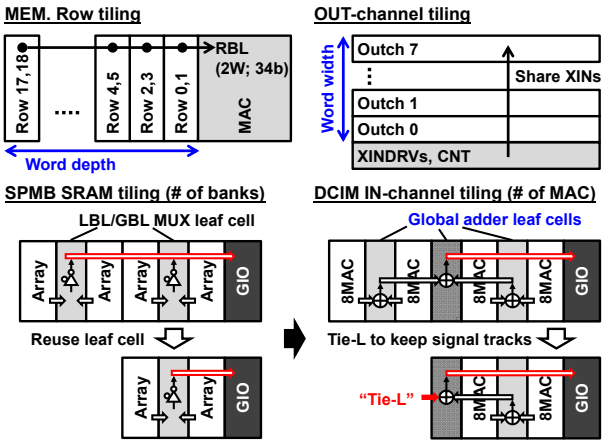


Fig. 7. DCIM compiler tiling support.

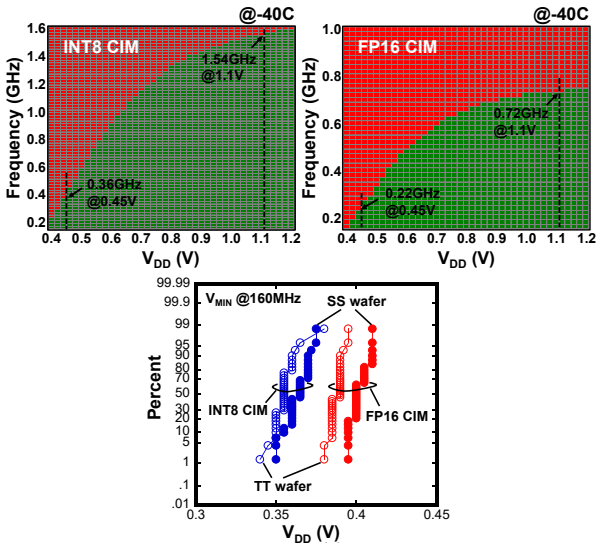


Fig. 8. V_{MIN} and shmoo plots for INT8 DCIM and FP16 DCIM.

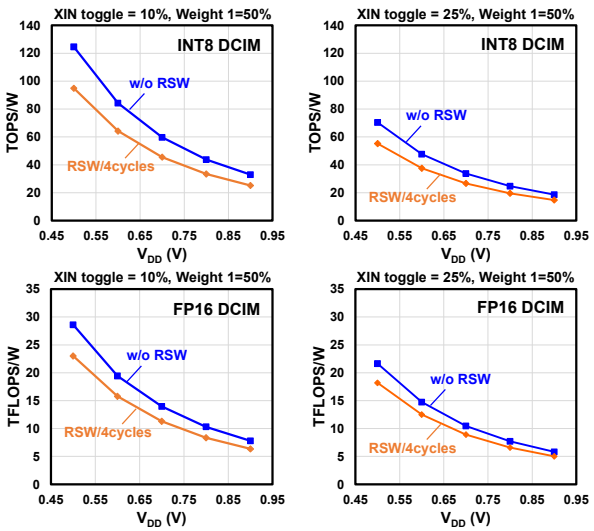
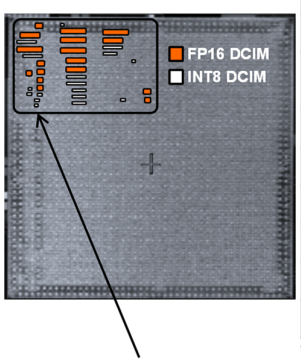


Fig. 9. Power comparison with voltage sweep, different XIN toggle rate, and different memory row switch frequency (RSW).

		ISSCC'24 [1]	ISSCC'23 [3]	ISSCC'23 [4]	This work
Technology		3nm	22nm	28nm	3nm
Bitcell		6T	6T	6T + FCU	LV 12T cell
VDD		0.36 ~ 1.1V	0.6 ~ 0.8V	0.6 ~ 0.9V	0.37 ~ 1.1V (INT8) 0.41 ~ 1.1V (FP16)
Compiler support		No	No	No	Yes
Input Ch range		72	128	128	64 ~ 16ch, step16
Output Ch range		4	32	8	8 ~ 2ch, step2
#rows range		18	8	8	18 ~ 2 rows, step2
INT	Macro area	0.0167 mm^2	N/A	0.146 mm^2	0.0175 mm^2 (*)
	Input format (W/XIN)	INT12/INT12		INT8/INT8 BF16/BF16	INT8/INT8
	Output format	INT31		INT23 or BF16	INT22
	TOPS/W	@0.55V 32.5 (XIN toggle 10%)		@0.6V 44 (XIN1=50%)	@0.5V (*) 124.6 (XIN toggle=10%) 70.4 (XIN toggle=25%)
	TOPS/mm^2 (**)	55.0 @0.9V		2.5 @0.9V	90.2 @1.1V (*)
FP	EXP alignment method	N/A	Alignment first	Alignment first	Multiply first
	Macro area		N/A	0.146 mm^2	0.0427 mm^2 (*)
	Input format (W/XIN)		BF16/BF16	INT8/INT8 BF16/BF16	INT8/INT8 FP16/FP16
	Output format		FP32	INT23 or BF16	INT22 or FP32
	TFLOPS/W		@0.6V, BF16 70.21 (XIN1=10%)	@0.6V, BF16 31.6 (XIN1=50%)	@0.5V, FP16 (*) 28.6 (XIN toggle=10%) 21.6 (XIN toggle=25%)
	TFLOPS/mm^2		N/A	2.0 @0.9V, BF16	17.4 @1.1V, FP16 (*)

(*) with biggest macro configuration (64in x 8out, 18rows).
(**) Normalized to INT8.

Fig. 10. PPA summary table and comparison to prior works.



Technology	3nm HKMG FinFET
Bitcell	LV 12T cell (0.055μm^2)
Macro size	355.5 μm x 49.2 μm (INT8) 464.1 μm x 92.1 μm (FP16)
Memory config.	153 Kb 576 addresses x 272 cols, Write & Read w/ MUX1
TOPS/mm^2 INT8@1.1V	90.2 TOPS/mm^2
TOPS/W (*) INT8@0.5V	124.6 (XIN toggle=10%) 70.4 (XIN toggle=25%)
TFLOPS/mm^2 FP16@1.1V	17.4 TFLOPS/mm^2
TFLOPS/W (*) FP16@0.5V	28.6 (XIN toggle=10%) 21.6 (XIN toggle=25%)
SQNR-based accuracy (dB)	MobileVit-v2-0.5: 51.0 Inception-v1: 53.2
Inference accuracy (%)	MobileVit-v2-0.5: 71.4 Inception-v1: 71.3

w/ biggest config. w/ 64in x 8out, 18rows.
(*) w/ W1 = 50%, w/o RSW.

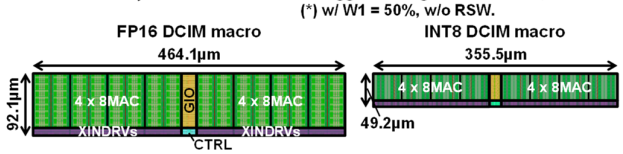


Fig. 11. Chip photograph and summary table.

References

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